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Geometric Analysis and YOLO Algorithm for Automatic Face Detection System in a Security Setting

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Abstract. Face detection is a computer technology that determines the location and size of the human face in an arbitrary digital image employed for authentication. Face recognition technology plays a vital role in network security, video compression, content indexing, and retrieval since "humans" are the focal point of many videos. Face recognition-based network access control makes it almost difficult for hackers to obtain a user's "password" and improves user-friendliness in human-computer interaction. In this paper, an automatic face detection system that accurately detects human faces and ignores any other object that is not a human face using geometric analysis and the you-only-look-once (YOLO) algorithm is introduced. The system is able to predict the ages and genders of the faces detected. It also detects the facial landmarks of the faces and indicates the emotions of the faces detected. A sample of four faces is considered for testing the system; thus, accurately detecting gender and emotion but not age correctly. In all, the evaluation shows about 80% accuracy. With the results got, the system can support security and analytics. It can be used to get analytics in an event with a lot of attendees and can also be used to get a facial mapping of someone involved in a crime scene for security purposes.

Keywords: Face detection, Geometric analysis, Gender estimation, Age estimation, Facial recognition

1.0 Introduction

Facial recognition is one activity performed daily by humans as a routine effortlessly [1]. The vast accessibility to embedded, low-cost and powerful computer systems has led to great interest in automated image processing. This has been applied in human-computer interaction, environment surveillance, multimedia management, and biometric authentication. The need to properly classify humans is growing [2]. But, there is no better way to achieve this than using facial recognition. Facial recognition offers many advantages over other biometric modalities such as iris and fingerprints. Outside the natural and nonintrusive importance, it offers the benefit of distant capturing in a covert form [3]. It is already gaining lots of interest from the research bodies and society [3]. Industries that consider people as the centre of focus, such as media, security, and finances, use this technology [4]. The use of facial recognition for controlling network access makes it difficult to steal passwords and fosters friendly interaction between

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computers and humans. The retrieval and indexing of video data on the detection of a particular person will aid users who are political scientists, news reporters, content creators, and moviegoers. It has been shown from research that faces speak and express emotions in different forms [5]. The major objective of this study is to design a system that accurately detects human faces and ignores any other object that is not a human face. The study's specific objectives are to: predict age of the faces detected, Predict the Gender of the faces detected, Detect Facial Landmarks of the faces detected, and predict Emotions of the faces detected.

2.0. Overview of Related Approaches

The computer technology used to determine the size and location of human faces from a random digital image is facial detection. The features of the face are detected alongside ignoring other objects seen. This is because the face(s) is/are the major consideration from the entire image. This can be considered a precise case of object class detection, where the major class identifies the locations and sizes of objects belonging to a particular class from an image. Detecting faces can be considered as a broad case of face localisation. Metze et al. [6] compared four different approaches to recognize the gender and age of people from telephone speech; these include a parallel phoneme recogniser system for comparing Viterbi decoding scores. Human beings perceive who a person is not only by their facial form but also by some other features, including hair form, dresses, skin tone, and whole body. Addressing the problem of gender classification using human faces has been studied extensively. There are two ways that are widely used. The first is suggested by [7], who use a Support Vector Machine (SVM) to classify people's genders using thumbnail facial photos. Baluja et al. [8] introduced the second, which used the Adaboost algorithm to predict gender.

The vast interest in the adoption of Local Binary Patterns (LBP) for facial led to the use of its histogram features in the representation of gender features in [9]. This was enhanced by using the Adaboost algorithm for learning the best local classification features. The authors experimented with making a prediction of the gender, age, and likely ethnic group of faces from images. For gender categorization, other local characteristics have also been used. Cheng et al. [10] proposed an approach for recognizing gender from images using shape contexts and Scale Invariant Feature Transform (SIFT) descriptors. The authors improved the method by using Adaboost for selecting features and classifying them from the face images. Agbo-Ajala and Viriri [11] conducted research to classify face-based gender from consumer images gotten from a multi-ethnic face database. To overcome the non-uniformity of a pose, illumination changes, and expression, Active Shape Models (ASM) was used to normalize the facial texture. The conclusion from work is that putting ethnic factors into consideration can foster the improvement on the accuracy of gender classification in a multi-ethnic environment. The LBP has been seen to prove good discrimination capability with simplicity compared to SIFT and HOG. To institute a baseline for appearance-based means, LBP and SVM were employed to predict gender from facial images in the research work.

Other methods for improving facial recognition performance include lexical features [12], frame and utterance-based acoustic-prosodic joint features [13-16], and novel acoustic features [17].

According to Ajmera and Burkhardt [17], the cepstral coefficients were subjected to a discrete cosine transform and cepstral trajectories in a bid to lower (3-14 Hz) modulation frequencies. A fuzzy membership is assigned as a weight to each training data point in the fuzzy SVM modeling approach suggested by Nguyen et al. [18] to boost robustness against noise and outliers.

In addition, techniques from language identification applications and speaker verification such as GMM-SVM mean super vector systems [19], nuisance attribute projection (NAP) [20], anchor models [20, 21],

and Maximum-Mutual Information (MMI) training have been successfully utilized in terms of speaker-age and gender [21].

An anchor model was used by employing a back-end SVM in [20]. This was used to distribute similarity scores amid the training dataset and all the anchor speaker models. From the literature, it can be deduced that combining the right classification methods often improves the system's overall performance significantly [14, 19, 21, 22, 23].

Considering the knowledge gotten from literature, a right combined method is essential for carrying out an efficient facial recognition system using age and gender estimation. This research does so by employing geometric analysis, and you only look once algorithms.

3.0 Research Methodology

The proposed system was designed using the agile Software Development Lifecycle (SDLC) model. This employs both incremental and iterative processes with the consideration of the adaptation of the process and the satisfaction of customers for the speedy delivery of the product. This technique breaks down the product into bit-by-bit incremental constructions. Such bits are provided in iterations.

3.1 Geometric Analysis

We explore geometric features in our study, considering the distances between one facial point and another. To improve the rate of recognition, a steady feature set is essential for changing the shooting conditions. The similarity of faces can be evaluated with some steps in the selection of faces "from coarse to fine." The first step is the detection of a set of fiducial points from all faces in an image or video, and selecting the close k_1 images close to the query face using some geometric features depending on the points, where k_1 may be closed to in equation 1:

$$K_1 = \left(lnN\right)^2 \tag{1}$$

When this is done, the images are rotated and scaled with the correlation evaluated between the central face parts of the k_1 images and the query image. At the end of carrying out this, k_2 images from the set of k_1 are selected in equation 2, where:

$$\ln\left(K_1\right) \le K_2 \le \left(\ln K_2\right)^2 \tag{2}$$

The final images will have at least one similar image of the person queried, if any.

The fiducial points have to do with regions from the specimen marked for aligning subsequent sections. The researcher may choose to put the fiducial points or trace fiducial contours on the image. The fiducial points in this research are automatically detected. The geometric features are shown using areas, perimeters, and segments using the detected points. Different subsets of the features were studied to find the most robust features. As shown in equation 3, a feature set can be optimized after being obtained using the technique. Here, feature space is identified using the maximal distances between the clusters and the minimal among the patterns of one cluster.

$$F = \sqrt{\frac{\sum_{i=1}^{k}\left(M_{D_i} - D_i\right)}{\sum_{i=1}^{k}\left(M_{M_i} - D_i\right)^2}} \tag{3}$$

where M_i and D_i are mean and variance of the feature values for k images of the *i*-th person, M_{Di} and M_{Mi} are mean of D_i and M_i, respectively. If the result provides the lowest F value, this means a better feature set has been identified. As a means of validating the proposed technique, when the feature space changes, the computation of the cluster distances is done. The predicted value p_c, the probability an object exist in a bounding box is also computed as can be seen in Figure 1.

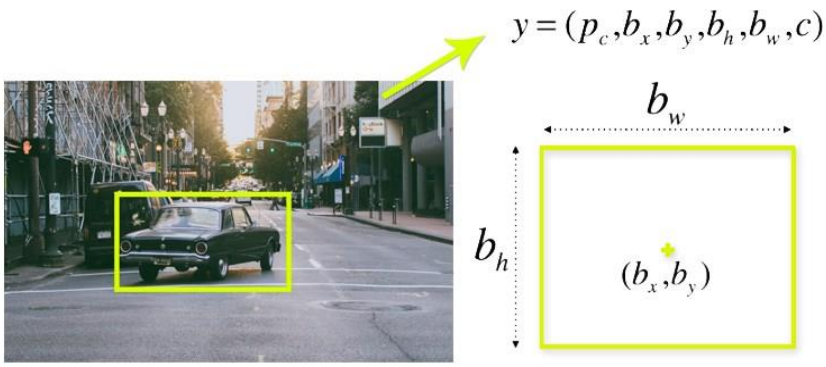

Figure 1. Fitting Bounding Boxes (Maj (2018))

The You Only Look Once algorithm does not search targeted regions from an image likely to contain a certain object. Its focus is to split out an image into cells, 19×19 grid. Each of the cells is responsible for predicting the 5 bounding boxes (possibly, more than one object is present in this cell). Hence, resulting to 1805 bounding boxes for a single image, as shown in Figure 2.

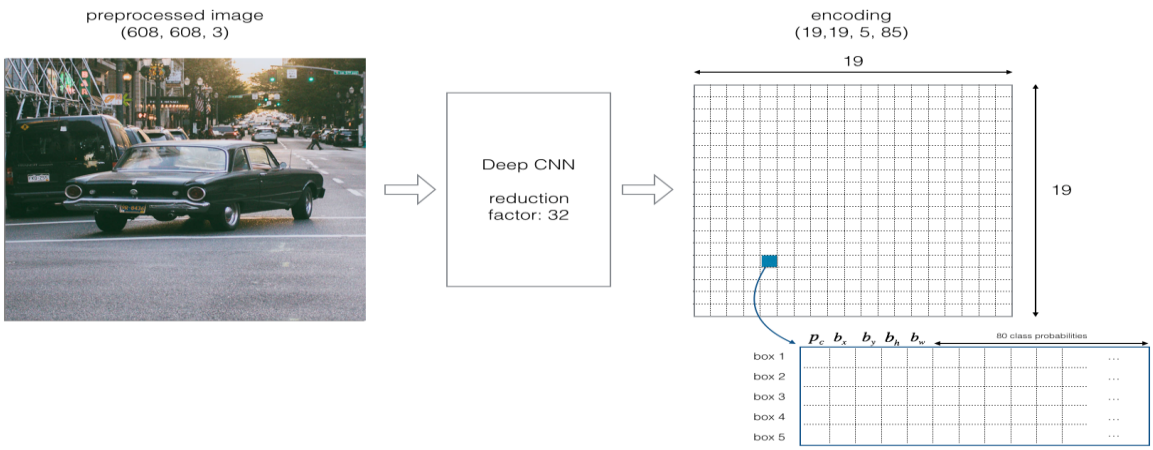

Figure 2. Image Encoding (Maj (2018))

Most of these cells and boxes would not have objects in them, which is the reason for prediction. The next phase removes boxes with low object probability and bounding boxes using the highest shared area. This process is known as non-max suppression, as seen in Figure 3.

Figure 3. Before and After non-max suppression (Maj (2018))

4.0` Results and Discussion

The research work was implemented using NodeJS for running the Javascript program on the back-end and the server-side rendering of web assets. Javascript is an object-oriented computer programming language that is extensively used in web browsers to produce interactive effects. The creation of basic elements such as buttons, text fields, dropdown menus etc., was done using HyperText Markup Language. To add styling to the HTML webpages, we made use of Cascading Style Sheet (CSS). It was used to define web page style, including the design, layout, and variations in display for different devices and screen sizes. The different phases of the Automatic Face Detection System using Gender and Age Estimation (AFDUG-SYS) are webcam face detection, webcam face expression detection, webcam age and gender recognition, video face tracking, and webcam face landmark detection.

The webcam face detection section activates the host machine webcam to detect human faces and put a bounding box on the face detected, as depicted in Figure 4. The webcam face expression detection section activates the host machine webcam to detect human faces and put a bounding box on the face along with the result of the human facial expression predicted on the bounding box as depicted in Figure 5. The webcam age and gender recognition section activate the host machine webcam to detect human faces and put a bounding box on the face along with the gender and age predicted on the bounding box as depicted in Figure 6. The video face tracking section uses a video from the host machine secondary memory to look for the face in the video; a bounding box is put on the faces detected, as depicted in Figure 7. The webcam face landmark detection section activates the host machine webcam to detect human faces and put a bounding box on the face along with lines drawn over facial landmark features such as the eyes, nostrils, and jaw, and mouth, as depicted in Figure 8.

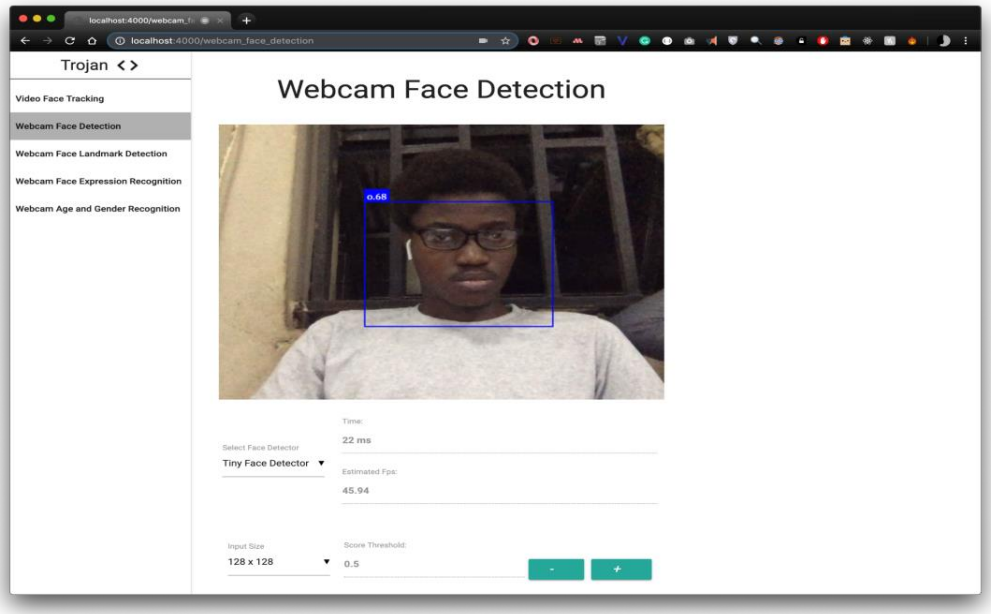

Figure 4. Webcam Face Detection

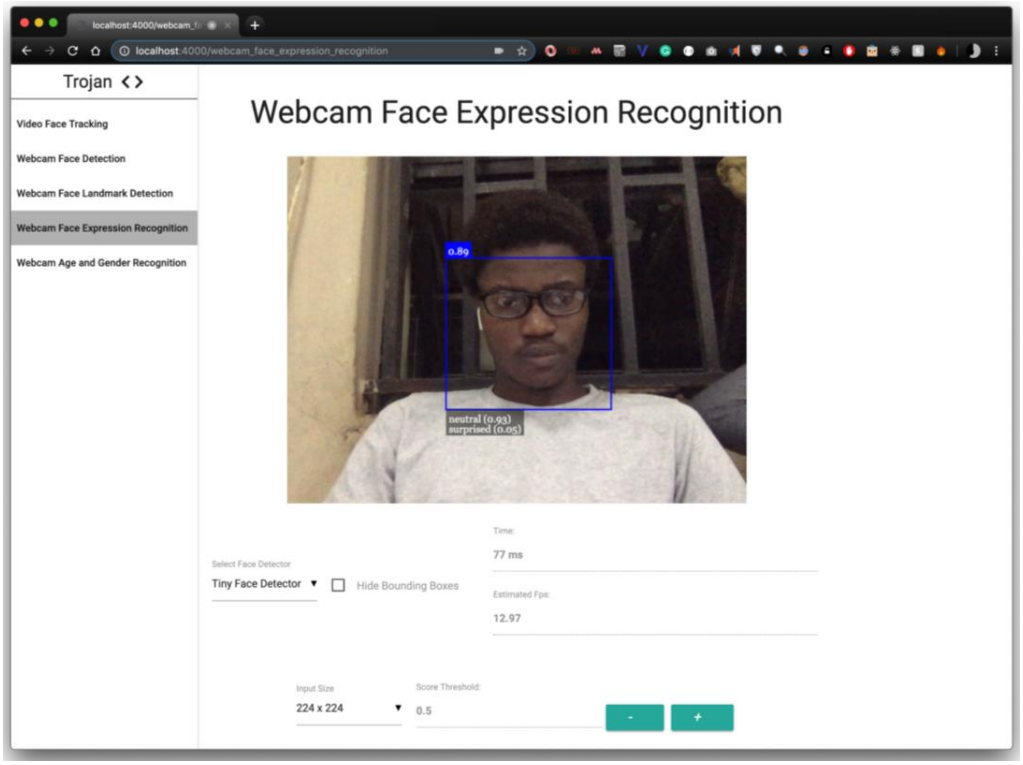

Figure 5. Webcam Face Expression Recognition

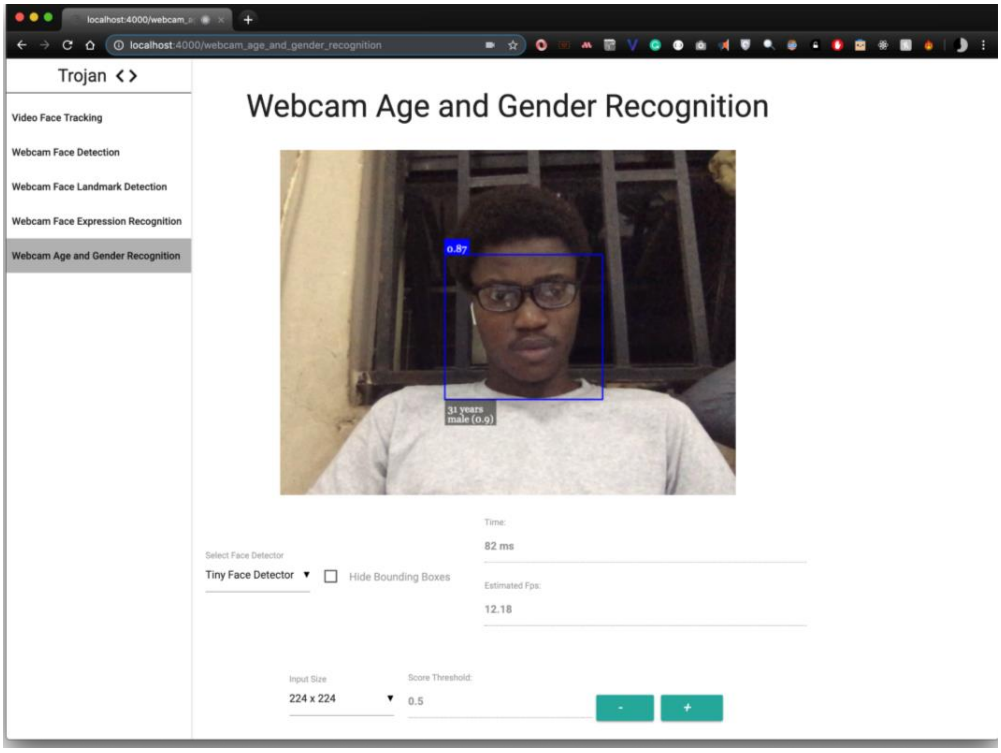

Figure 6. Webcam Age and Gender Recognition

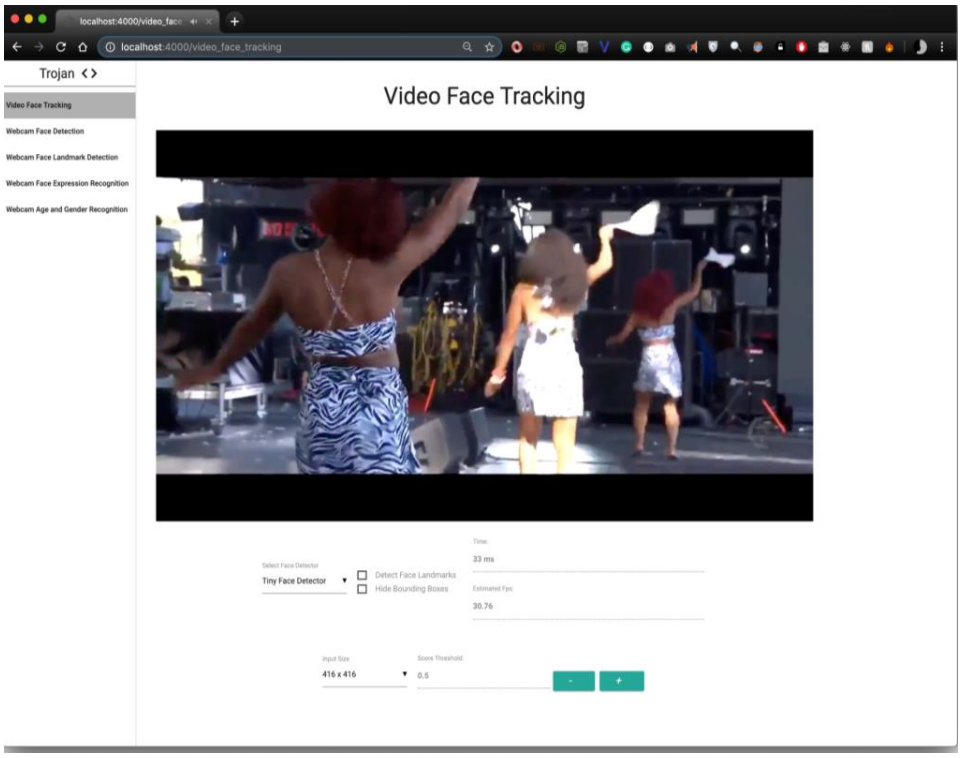

Figure 7. Video Face Tracking

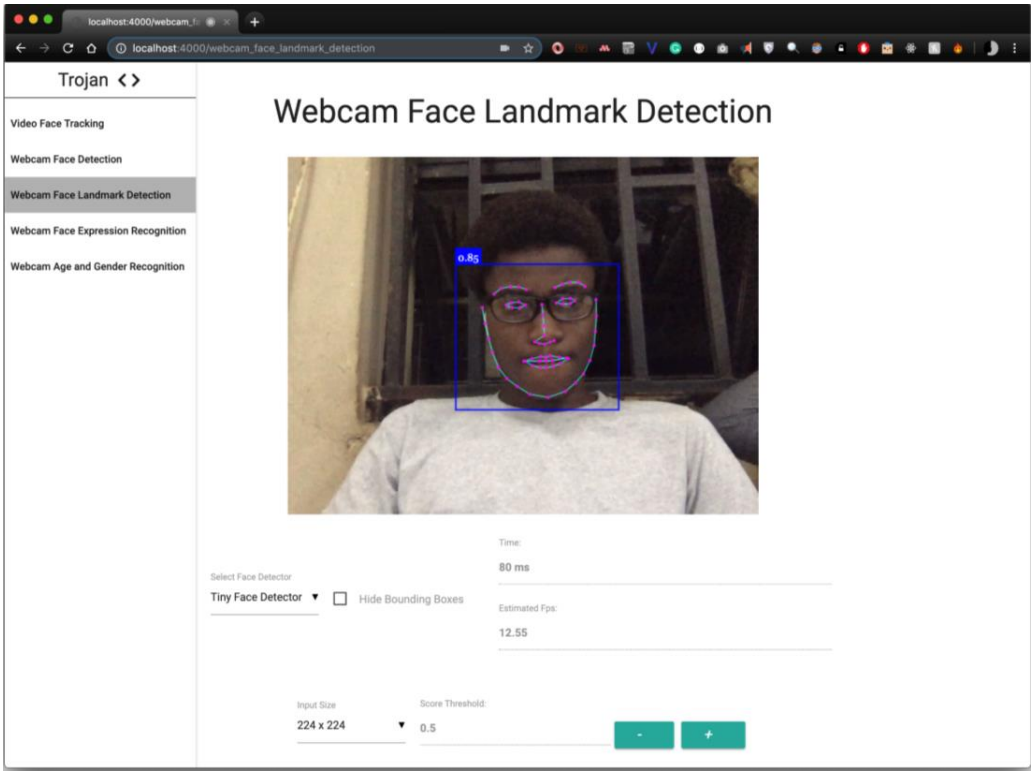

Figure 8. Webcam Face Landmark Detection

4.1 Evaluation and Results

This system was tested on four McPherson University Students, two males and two females. The aim of the test is to evaluate if the system can predict the right Gender, Age, and Facial Expression. As shown in Table 1, we collated the age and gender of the subjects and exposed them to the system to see if it accurately predicted the attributes mentioned earlier.

Table 1. Result from testing AFDUG-SYS

Student Name	Facial Expression	Age	Predicted Expression	Predicted Age	Gender	Predicted Gender
Mary	Happy	21	Happy	25	Female	Female
Zara	Happy	23	Happy	23	Female	Female
Wonderful	Neutral	20	Neutral	22	Male	Male
Trojan	Sad	19	Sad	19	Male	Male

In Table 1. We can see that the system got the right gender for every subject and facial expression, but it is not so accurate when it comes to age prediction. Though it accurately predicted the ages of two subjects, the remaining two candidates' age predictions were off by +2 to +5. Age prediction is subject to lighting conditions to be able to pick up skin tone and facial features that may appear old or young, as well as training data the system was subjected to.

5.0 Conclusion

The study concludes from the results obtained that using YOLO with Geometric analysis can accurately detect the gender of a face and predict age. This made geometric analysis a good method for facial detection and feature prediction. A face detection system with an age and gender prediction model was designed to detect human faces, detect/predict the gender and estimate the age of the human face. This system was designed with the sole aim of security and analytics. It can be used to get analytics in an event with a lot of attendees and can also be used to get a facial mapping of someone involved in a crime scene for security purposes. This system implemented the use of YOLO to quickly and accurately detect human faces while using geometric analysis to map the human face. YOLO was initially designed for object detection, but it was re-trained to further detect human faces to improve the system's accuracy and speed. Using HD (High Definition) cameras and computers with high processing power can analyse the realm video footage. The system can be further implemented for capturing a number of participants in an event or used as an attendance management system in schools.

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